

Further Maths etc. — document structure

A map of the single self-contained `index.html` file: how the file is laid out, what each region does, and where to edit. Line numbers refer to the current file and will drift as you add content, but the ordering and section markers stay put.

1 · The file at a glance

One HTML file, no build step, no external dependencies beyond a Google Fonts link. It splits cleanly into three concerns, which is the whole point of the architecture: you almost always only touch the third.

	Region	Lines	What it is
A	<head> + styles	1–169	Meta, font link, and the entire CSS stylesheet (design tokens + all component styling).
B	<body> markup	171–240	The static page skeleton — nav, hero, section shells, the detail panel, footer. Mostly empty containers that JS fills.
C	<script>	242–565	Your content data object, then the rendering engine that turns it into the page.

The three-layer rule. Shared base styles and the swappable skin (the colour/look tokens) are deliberately kept apart from the content data. Day-to-day you edit the **PAPERS** data block (layer C); restyling means touching the tokens (layer A); the rendering engine in between is designed to be left alone.

2 · Inside the stylesheet (lines 10–169)

The CSS is organised into commented banners, each marked with a `/* ===== NAME */` divider so you can jump straight to a component.

Lines	Banner	Controls
12–23	TOKENS (:root)	All design variables: backgrounds, ink colours, grid line, the five paper accent colours, fonts, max width, corner radius. This is the skin.
24–31	base	Box-sizing, body background + the 32px squared-paper grid, headings, the .wrap container, eyebrow label.
33–46	NAV	Sticky translucent header, brand wordmark, the P1–P4 + etc. pill buttons and their per-paper hover colours.
48–63	HERO	Big title, the italic purple “etc.” link, the wiggling “s” keyframes, lead text, the decorative hero curve, stat row.
65–71	SECTION	Generic section padding and the section heading row with its right-aligned hint.
73–91	PAPER CARDS	The 4-up responsive card grid, card hover lift, accent bar, and per-paper accent colour wiring.
93–103	ENRICHMENT (home)	The etc. grid on the landing page, item cards, and the coloured tag chips.
105–155	DETAIL	The per-paper view: back link, detail header, topic filter chips, the accordion topic items + resource rows, and past-paper rows.
157–168	FOOTER + motion	Footer, the SVG “draw” line animation, and the prefers-reduced-motion overrides.

3 · The body skeleton (lines 171–240)

Almost every visible element is an empty container with an `id`; the script writes into it on load. Knowing these `ids` tells you exactly where each piece of content lands.

Lines	Element	Role
173–184	<code><header class="nav"></code>	Sticky nav with the brand and five pill buttons (<code>data-p = p1...p4</code> , <code>enr</code>).
187–217	<code><main id="home"></code>	The landing page. Holds the hero, the four-paper section, and the etc. section.
188–196	<code>· section.hero</code>	Title markup is static; <code>#hero-lead</code> , <code>#hero-stats</code> and <code>#hero-curve</code> are filled by JS.
198–206	<code>· #papers</code>	Empty grid; the four paper cards are injected here.
208–216	<code>· #enrichment / #enr-grid</code>	The etc. section; collated etc. items are injected into <code>#enr-grid</code> .
219–233	<code><section id="detail"></code>	Hidden by default; shown when a paper opens. <code>#d-pn</code> , <code>#d-title</code> , <code>#d-codes</code> , <code>#d-blurb</code> , <code>#d-curve</code> and <code>#d-body</code> are populated per paper.
235–240	<code><footer></code>	<code>#footer-note</code> and <code>#footer-meta</code> filled from the <code>COURSE</code> object.

4 · The script (lines 242–565)

Two halves, divided by a banner comment. The top half is yours to edit; the bottom half renders it.

4.1 — Content data (the part you edit)

Lines	Object	What it holds
263–267	COURSE	Teacher name, hero lead paragraph, footer text. Three strings.
269–366	PAPERS[]	An array of four paper objects (<code>p1–p4</code>). Each has: <code>id</code> , <code>n</code> , <code>name</code> , <code>curve</code> , <code>codes</code> , <code>blurb</code> , a <code>topics[]</code> list, and five content arrays — <code>slides</code> , <code>textbook</code> , <code>videos</code> , <code>enrichment</code> , <code>pastpapers</code> .

How an item files itself. Each entry in `slides / textbook / videos / enrichment` carries a `topic` string that must match a name in that paper's `topics[]` array exactly — that match is what slots it under the right syllabus section. A url of `"#"` (or omitted) renders as a placeholder and won't open; a real path opens in a new tab. `isNew:true` adds a NEW badge.

4.2 — Rendering engine (leave alone)

Lines	Piece	Job
371–373	helpers	<code>\$</code> selector shorthand, <code>av()</code> accent-var lookup, <code>shortName()</code> for paper labels.
375–380	<code>ALL_ENR</code>	Collates every paper's enrichment into one list for the home etc. grid — authored once, shown in both places.
382–419	curve SVGs	<code>curveSVG()</code> draws each paper's maths signature: rose (<code>P1</code>), 7th-roots heptagon (<code>P2</code>), parabola (<code>P3</code>), normal curve (<code>P4</code>), lattice fallback.
421–431	hero fill	Tally helpers plus writing the lead text, hero curve, and the five stat counters.
433–456	card + etc. render	Builds the four paper cards (<code>#papers</code>) and the collated etc. grid (<code>#enr-grid</code>).
458–545	<code>openPaper()</code>	The big one: builds a paper's detail view — topic chips, the per-topic accordion of resource rows, and the past-paper block — then wires up the click handlers.
546–564	routing	<code>closePaper()</code> , <code>route()</code> reading the URL hash (<code>#p1</code> , <code>#p1~2</code> , <code>#enrichment</code>), card/pill click handlers, footer text, and the initial <code>route()</code> call.

5 · How content flows to the screen

1. You add an item to a paper's array in **PAPERS** (layer C), tagging it with a matching topic.
2. On load, the engine tallies everything and renders the hero stats and the four paper cards.
3. Any enrichment entries are also collated into **ALL_ENR** and shown on the home etc. grid, deep-linked back to their section.
4. Clicking a card (or a pill, or an etc. link) sets the URL hash; `route()` reads it and calls `openPaper()`.
5. `openPaper()` groups that paper's items by topic into the accordion, plus the past-paper rows if any sessions exist.

6 · Where to edit — quick reference

I want to...	Go to
Add a solution / video / slide / etc. piece	The relevant array inside that paper in PAPERS ($\approx$ 269–366)
Add a past-paper session	That paper's <code>pastpapers:[]</code> array
Change the page background colour	--bg, line 13
Change or remove the grid lines	--grid (line 15) for colour; line 27 for the grid itself
Recolour a paper accent	--p1...--p4 / --enr, lines 16–20
Edit hero lead / footer text	The COURSE object, lines 263–267
Change a paper's name, codes, or blurb	That paper's header fields in PAPERS

Generated as a structure reference for `index.html`. Line numbers reflect the file at the time of writing.